

Ankit Kumar

M.Tech, Department of Management Sciences (IME)
Indian Institute of Technology, Kanpur

Email:ankitk25@iitk.ac.in | **LinkedIn:** Ankit Kumar
Github: Ankitiitk369 | **Portfolio:** Ankit Kumar

Academic Qualifications			
Year	Degree/Certificate	Institute	CPI/%
2025 - Present	M.Tech (DoMS)	Indian Institute of Technology, Kanpur	7.18/10
2021 - 2025	B.Tech	C.S.A. University of Agriculture and Technology, Kanpur	8.25/10
2020-2021	UP Board (XII)	Government Inter College Kadaura, Jalaun	77.20%
2018-2019	UP Board (X)	Government Inter College Kadaura, Jalaun	80.83%

Professional Experience

Tata Consultancy Services (TCS) | Data Science & AI Intern | Computer Vision & ML

(May'26 - July'26)

- Engineered an automated end-to-end Vehicle CO2 Emission Tracking & Intelligence pipeline from traffic CCTV feeds to monitor urban carbon footprints and support ESG reporting and sustainability analytics
- Implemented YOLOv8 deep learning localization with an OpenCV preprocessing engine (Bilateral Filtering, CLAHE, Otsu Binarization) and EasyOCR, paired with a custom Indian plate regex syntax corrector (resolving optical confusions like O vs 0, I vs 1, B vs 8) to enhance OCR character recognition on degraded video /image frames.
- Integrated Vahan / RTO registry APIs & offline databases across 36 Indian States/UTs to dynamically extract vehicle metadata (Fuel Type, Engine CC, Vehicle Class) and compute dynamic Vehicle Age from Manufacturing Year.
- Modeled multi-vehicle trip trajectories using a Multi-Origin Convergence Routing approach with Haversine and OSRM distance matrices, enabling route-level distance estimation and fleet-level carbon-density analysis.
- Trained dual-stage ensemble ML models, where Gradient Boosting Regressor achieved R2 = 0.996 (RMSE =3.12 g/km) for emission rate prediction and Random Forest Regressor achieved R2 = 0.989 for total trip-emissions Prediction (kg), benchmarked against ARAI/IPCC emission standards.
- Delivered an interactive Streamlit Web Dashboard & automated multi-sheet Excel reporting pipeline, with live bounding box annotations, GIS convergence maps, and a 17-test Pytest validation suite with 100% pass rate

Automation of Asset Receiving Process Module in Global Procurement Web Application System

- Contributed as a team member to the Asset Receiving Automation Module for Global Procurement System, focused on digitizing the manual asset-receiving process.
- Assisted the frontend development team in creating Angular-based interfaces for capturing asset, Purchase Order, quantity, condition, and location details.
- Supported implementation of frontend validation and UI workflows, including PO validation, form validation, authentication, and role-based access.
- Assisted in developing asset-tracking interfaces such as receipt lists, filtering, pagination, image upload, and dashboard components

Key Projects

Zepto E-Commerce Data Analysis | SQL | Exploratory Data Analysis

(Self Project, Aug'25 - Sep'25)

- Investigated 3,700+ Zepto product records across 14 categories, identifying pricing, discount, and inventory trends.
- Performed data cleaning by removing invalid zero-price records and converting paise to INR for accurate analysis.
- Applied optimized SQL queries using SELECT, WHERE, GROUP BY, HAVING, CASE, and aggregations for business analysis.
- Generated insights on discounts, stockouts, revenue, inventory weight, and price-per-gram to support pricing decisions.

Customer Shopping Behavior Analysis | SQL | Power BI

(Self Project, Sep'25 - Oct'25)

- Analyzed 3,900 customer records with 18 attributes using Python and Pandas to identify retail behavior patterns.
- Cleaned 37 missing ratings and engineered age groups, spending tiers, and purchase frequency features for segmentation.
- Applied PostgreSQL queries using CTEs, window functions, subqueries, CASE, and aggregations for business analysis.
- Built an interactive Power BI dashboard with DAX measures, KPIs, slicers, and drill-throughs for customer insights.

Loan Portfolio Risk Analysis & Visualization | SQL | Tableau

(Self Project, Sep'25-Oct'25)

- Analyzed 38.58K loan applications to evaluate funding amounts, repayments, interest rates, DTI, and portfolio risk.
- Applied SQL queries using GROUP BY, CASE, aggregations, and filtering to segment loan portfolios by loan status.
- Identified 86.18% good loans and 13.82% bad loans, delivering critical insights to support credit risk assessment.
- Built interactive Tableau dashboards tracking loan applications, funded amounts, and monthly performance trends.

Credit Risk Modelling: Loan Eligibility Prediction | Classification

(Guide: Prof. Faiz Hamid, Aug'25 - Nov'25)

- Built an end-to-end credit risk model using 51,336 customer records and 79 features to predict loan eligibility.
- Performed feature engineering and statistical hypothesis testing using Chi-Square, ANOVA, VIF, and SMOTE for data.
- Developed robust ML pipelines using Logistic Regression, Decision Tree, SVC, Random Forest, and XGBoost classifiers.
- Optimized XGBoost with GridSearchCV tuning, achieving 78% accuracy for loan eligibility prediction on test data.

BigMart Sales Analysis Using Multivariate Regression | Causal Analysis

(Self Project, Aug'25 - Sep'25)

- Analyzed 8,523 BigMart sales records using Pandas, Seaborn, and Matplotlib to identify key product sales drivers.
- Performed data cleaning and feature engineering using grouped imputation, encoding, and outlet-age features for modeling.
- Applied OLS and Linear Regression with log transformation and VIF analysis, achieving a solid 72.1% R2 score on data.
- Selected four key predictors including Item MRP and outlet types, achieving 0.290 cross-validation MSE on sales data.

AcademicPath: Career & Course Guidance System | Multimodal RAG

(Self Project, Mar'26 - May'26)

- Developed a multimodal RAG assistant using LlamaIndex, Qdrant, Hugging Face embeddings, and Groq LLMs for guidance.
- Implemented semantic vector retrieval with course metadata and page citations across 113+ academic college courses.
- Built course-comparison routing and conversational memory to generate structured guidance for multi-course queries.
- Integrated Faster-Whisper and gTTS for real-time voice interaction, achieving 100% Context Recall and 70.1% Accuracy.

MediMind AI: Clinical Decision Support Platform | GenAI | RAG | LLMOps (Self Project, Mar'26 - May'26)

- Built an end-to-end RAG platform using **LangChain**, **Groq API**, and **Pinecone**, reducing query inference latency by **85%**.
- Implemented **MiniLM embeddings** and **Cosine Similarity**, achieving **96.8% RAG faithfulness** with clear source citations.
- Developed lab report analysis using **PyPDF**, detecting biomarker anomalies and exporting structured PDF clinical briefs.
- Integrated 4 risk calculators and **Plotly visualizers** with sub-10ms emergency triage guardrails for clinical decisions.

Dental Appointment Management System | Agentic AI | LangGraph (Self Project, June'26 - Aug'26)

- Developed conversational AI for dental appointment management, enabling **natural-language booking and rescheduling**.
- Implemented **LLM tool calling** with **LangGraph** and **LangChain** for live appointment search and MySQL database updates.
- Integrated structured appointment **dataset of 4,280 slots across 10 doctors and 7 specializations** for dental clinics.
- Built an **interactive LangGraph workflow** with state management and conditional tool execution for dental bookings.

Text-to-SQL & Agentic SQL Analytics System | LangGraph (Self Project, May'26 - Aug'26)

- Developed a **Text-to-SQL** system using **Gemini 2.0 Flash**, **LangChain**, and **MySQL** for natural-language SQL query tasks.
- Built an **agentic LangGraph ReAct** workflow using **SQLDatabaseToolkit** for schema inspection, query generation, and execution.
- Integrated **SQL query-checking tools** to iteratively validate queries and generate answers across enterprise data.
- Evaluated the pipeline using **RAGAS**, achieving **1.00 Context Precision** and **3.8/5 Helpfulness** across benchmark queries.

Air India Domain-Specific GenAI Knowledge Assistant | RAG | LLM (Self Project, Jan'26 - Mar'26)

- Built a domain-specific **AI assistant** using **LangChain**, **Groq Llama 3.3 70B**, and **Streamlit** for Air India document Q&A.
- Processed 121 PDF pages using **DirectoryLoader** and **PyPDFLoader**, with cached loading for fast application startup.
- Implemented **context-grounded prompting** with strict anti-hallucination guardrails, restricting answers to PDF docs.
- Integrated **dynamic LLM routing** between **Groq** and **xAI**, with **real-time token streaming** and user message chat history.

Solar Panel Defect Classification | Deep Learning | Computer Vision (Guide: Prof. Veena Bansal, Jan'26 - May'26)

- Developed a 6-class **CNN system** using **TensorFlow** and **transfer learning** to classify 885 solar-panel field image sets.
- Applied image **augmentation**, **Batch Normalization**, **Dropout**, and **L2 regularization** for superior model generalization.
- Benchmarked **CNN**, **MobileNetV2**, and **EfficientNetB0**, achieving **82.5% validation accuracy** after Keras Tuner optimization.
- Deployed optimized model using **Streamlit**, providing confidence scores, top-3 predictions, and probability plots.

Intelligent Document Processing for Invoice Extraction | OCR (Self Project, Feb'26 - Mar'26)

- Built an invoice processing pipeline using **Python**, **pdfplumber**, **PyMuPDF**, **Tesseract OCR**, and **OpenCV** for PDFs data.
- Processed 1,007 invoice PDFs with 100% pipeline success, extracting structured fields, tables, and invoice line-items.
- Implemented validation and confidence scoring, achieving **98.6% pass rate** and **0.948 average field confidence score**.
- Developed **FastAPI** and **Streamlit apps** processing invoices in **0.13 seconds**, generating JSON, CSV, and RAG outputs.

Marketing Campaign Performance Analysis | Statistical Testing | Regression (Self Project, Sep'25 - Oct'25)

- Evaluated Facebook and AdWords campaigns using **EDA** to compare clicks, conversions, costs, and marketing performance.
- Performed **A/B testing** using Welch's t-test, proving Facebook conversions were significantly higher than AdWords.
- Built **Linear Regression** to model click-to-conversion relationships, achieving a strong **R2 score of 76.35%** overall.
- Applied **cointegration testing**, confirming significant long-term equilibrium between digital ad costs and conversions.

Sentiment Analysis of Amazon Product Reviews Using Hybrid AI Techniques (Self Project, Oct'25 - Nov'25)

- Analyzed 11,000 Amazon reviews using **NLTK**, **tokenization**, **lemmatization**, and **TF-IDF** for product sentiment classification.
- Compared **Logistic Regression**, **Random Forest**, and **Naive Bayes**, with Logistic Regression achieving **85.4% accuracy**.
- Developed **RNN**, **LSTM**, **Bi-RNN**, and **Bi-LSTM models** using **Word2Vec**, with LSTM achieving **85.6% accuracy** on review text.
- **Fine-tuned BERT** for sentiment classification, achieving **90.4% accuracy** and **90% F1-score**, outperforming baselines.

Multi-Agent Resume Analyzer & Job Matcher | GenAI | CrewAI (Self Project, Jan'26 - Feb'26)

- Built a multi-agent resume analyzer using **LLMs**, **CrewAI architecture**, and **sequential workflows** for resume evaluation.
- Developed market research and SWOT agents using **Serper API** and **web scraping** to identify role-specific industry skills.
- Implemented resume-job matching using **TF-IDF**, **cosine similarity**, and **technical keyword extraction** for candidate skills.
- Built a **Streamlit dashboard** displaying match scores, skill gaps, SWOT analysis, and actionable career recommendations.

Electricity Demand Forecasting | Time Series | Deep Learning (Self Project, Aug'25 - Oct'25)

- Forecasted **monthly electricity demand** using statistical and deep learning approaches to support utility planning.
- Conducted time series **decomposition**; tested stationarity using **ADF test** and first-order series data differencing.
- Applied **ARIMA**, **SARIMA**, **Prophet**, and **LSTM models** using **ACF/PACF analysis** to capture seasonal demand trend patterns.
- Achieved the lowest **MAPE (2.46%)** on test data using **LSTM**, forecasting electricity demand for upcoming **36 months**.

Petrol Price Forecasting & Analytics System | Time Series | Machine Learning (Guide: Prof. Faiz Hamid, Aug'25 - Nov'25)

- Built a time-series forecasting system using **ARIMA**, **XGBoost**, **PyTorch LSTM**, and **Naive Baseline** models for fuel prices.
- Engineered 15+ temporal features including **lags**, **SMA**, **EMA**, **volatility**, **returns**, and **calendar variables** for fuel prices.
- Benchmarked models on 35% unseen data, with **XGBoost** achieving **1.12% MAPE** and 67.99% directional movement accuracy.
- Developed **Streamlit dashboard** and **FastAPI REST API** with **Plotly plots** and model comparison suites for forecasting.

Speech Sentiment Analysis using RNN & LSTM | NLP | Deep Learning (Guide: Prof. Veena Bansal, Jan'26 - May'26)

- Developed a Speech-to-Sentiment pipeline using **SpeechRecognition**, **GloVe**, and **Keras LSTM** for audio classification.
- Processed 14,402 airline tweets using text preprocessing and 100-dimensional GloVe embeddings for model training.
- Built a 2-layer **LSTM network** with 128 hidden units and **Dropout**, achieving **77.20% test accuracy** on airline tweets.
- Integrated **SoundDevice** recording and speech-to-text conversion for real-time sentiment prediction across 3 classes.

McDonald's Customer Segmentation & Marketing Analytics | Clustering | Machine Learning (Self Project, Sep'25 - Nov'25)

- Analyzed 1,453 customer survey records across 11 perception attributes to identify behavioral segments and personas.
- Benchmarked **K-Means**, **K-Modes**, and **Latent Class Analysis** using **StandardScaler**, **PCA**, and **correlation analysis** methods.
- Optimized K-Means with 100-fold bootstrap validation, achieving **0.592 Silhouette Score** and **0.930 Mean ARI** at K=5.
- Profiled five customer personas using demographics and behavior, translating segments into marketing strategies.

Sentiment-Based Product Recommendation System | NLP | Recommender (Self Project, Dec’25 - Jan’26)

- Analyzed 30K product reviews using **NLP preprocessing, lemmatization, TF-IDF, and SMOTE** for balanced classification.
- Benchmarked **Logistic Regression, Random Forest, and XGBoost**, with **Random Forest achieving 91% classification accuracy**.
- Developed User-User and Item-Item collaborative filtering using adjusted **cosine similarity**, achieving 2.55 RMSE.
- Built a hybrid recommendation pipeline combining Top-20 candidates and sentiment reranking for personalized Top-5 outputs.

NVIDIA Stock Price Prediction | Time Series Analysis | Quantitative Analysis (Self Project, Dec’25 - Jan’26)

- Developed forecasting models for **Nvidia stock prices** using 15 years of data with statistical and ML methodologies.
- Conducted **data visualization** and examined stationarity, seasonality and trends using **ADF, ACF and PACF plots** daily.
- Implemented statistical models: **ARMA, ARIMA, and SARIMA**, along with **Deep Learning (LSTM)** for stock price sequences.
- **LSTM** outperformed all models with **RMSE = 11.4**, followed by **SARIMA (RMSE = 21.3)** on out-of-sample stock market data.

AI-Powered Gen Z Consumer Behavior & Marketing Intelligence | MR project (Guide: Prof. Giridhar, Mar’26 - May’26)

- Analyzed **5,203 consumer records**, engineering **41 behavioral and demographic features** for consumer purchasing analysis.
- Applied **Chi-Square and Cramer’s V**, identifying significant FOMO and income-spending relationships with $p < 0.0001$.
- Benchmarked 6 classification models using **Stratified 5-Fold CV**, with tuned **CatBoost achieving 67.5% accuracy** rate.
- Applied **TreeSHAP and K-Means segmentation**, identifying key purchase drivers and 4 distinct consumer persona profiles.

Data-Driven Supply Chain Solution | Supply Chain Analytics | ML (Self Project, Feb’26-Apr’26)

- Analyzed diverse **supply chain data** for **100 SKUs**, effectively identifying inefficiencies in product and transport route.
- Performed dynamic **EOQ optimization**, thoroughly assessed potential **risks**, and evaluated inventory **KPIs** for warehouse.
- Applied robust **LightGBM (37% RMSE)** for sales forecasting and optimized **KNN algorithms** for vehicle route delivery.
- Delivered data-driven **insights** improving **stock levels**, reducing transport cost, and boosting forecasting accuracy.

Cancer Risk Level Prediction | XGBoost | Multiclass Classification (Self Project, July’26-Aug’26)

- Analyzed 2,000 patient records across **demographic, lifestyle, environmental, and genetic factors** for **multiclass risk classification**.
- Identified and removed target leakage from **Overall Risk Score**,addressed **class imbalance** using **SMOTE** and **class-weighted learning**
- Optimized **XGBoost with Optuna** using 40 trials and 3-fold **stratified cross-validation** for multiclass risk prediction..
- Achieved **88 % test accuracy, 0.72 Macro-F1, and 0.87 weighted F1**, deploying the prediction system through **Streamlit**.

Coursework & Skills	
Courses	Probability & Statistics Data Mining and Knowledge Discovery Introduction to Computing Operations Research for Management Applied Machine Learning (AML) Statistical Modelling for Business Analytics(SMBA) Business Process Management Supply Chain Management
Online Courses	Python for Beginners – Udemy Python Machine Learning: From Beginner to Pro – Udemy Power BI Workshop – Office Master Geodata Processing using Python and Machine Learning – IIRS (ISRO) Deloitte Australia Data Analytics Job Simulation – Forge British Airways Data Science Job Simulation – Forge Oracle Cloud Infrastructure 2025 Certified Data Science Professional – Oracle University Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate – Oracle University
Technical Skills	Python, SQL, Excel, Power BI, Tableau Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, TensorFlow, PyTorch GenAI, LangChain,Ollama, LangGraph, LlamaIndex, Qdrant, Pinecone, Streamlit, FastAPI, Github, Autoencoders, Transfer Learning,
Soft Skills	Team Management, Leadership, Decision Making, Communication Skills, Adaptability, Problem Solving,Teamwork, Critical Thinking

Positions of Responsibility

- **Alumni and Corporate Relations Coordinator, IIT Kanpur:** Actively liaising with alumni network and corporate partners to coordinate guest lectures, industrial collaborations, and departmental placement engagements.
- **Campus Coordinator, Mimamsa 2025 (IISER Pune):** Spearheaded institutional outreach across northern universities, managing registration workflows and logistical execution for national-level science competition.
- **NSS Student Volunteer (2 Years):** Organized extensive grassroots campaigns including Girls Education advocacy, digital literacy drives, and poverty surveys under the “*Ek Bharat Shreshtha Bharat*” initiative

Scholastic Achievements

- Successfully completed **GCI World April 2026 in Data Science & AI**, organized by **Matsuo-Iwasawa Laboratory, The University of Tokyo**, and passed the **final assessment**.
- Awarded the **Silver Medal Award** for **outstanding academic performance** in B.Tech by the **Chancellor**.
- Qualified the **SEBI–Investor Certification Examination in 2025** conducted by **NISM**